Possible Problems for Quiz 11 to be taken 4/17/19. Of these problems I will choose 3 to be on the quiz.

Problem 1

State the definition of a linear transformation

Problem 2

Determine if $T: R^{2\times 2} \to P^2$ given by $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a-d)x^2 - bx + c$ is a linear transformation

Problem 3

Describe the Kernel and range of the linear transformation taking $T: C[-1,1] \to R$ via $\int_{-1}^{1} f(x)dx$.

Problem 4

If $T\big(f(x)\big) = 1, T\big(g(x)\big) = 2, T\big(h(x)\big) = 5$

Find $T(4f(x) - 2g(x) + h(x))$